Akhil Kumar Puri

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PROFESSIONAL SUMMARY

Software Engineer with 4 years of experience building and maintaining backend services and full stack applications across fintech, healthcare, and enterprise SaaS. Reduced p95 read latency 32% at MUFG, cut lab results portal load time 63% at Quest Diagnostics, and consistently shipped production fixes in regulated environments with strict change management. Java 17-21, Python, Spring Boot 3.x, Kafka, React, Angular, GraphQL, PostgreSQL, AWS EKS, Azure AKS. M.S. Computer Science, Rowan University, Dec 2025.

EXPERIENCE

MUFG | Jersey City, NJ | Fintech | Dec 2025 - Present

Software Engineer - Full Stack

- Traced a silent data corruption defect in the transaction enrichment service where unrecognized international currency codes were defaulting to zero instead of failing loudly, skewing behavioral features fed to downstream fraud scoring models on a subset of wire transactions for several weeks. Root cause was a legacy internal currency code table maintained by a separate team that predated the current service. Built a reconciliation pipeline routing unrecognized codes to a manual review queue and patched the silent default.
- Extended Java 21/Spring Boot 3.x fraud detection microservices to support a new anomaly scoring rule set using Factory and Strategy patterns; reduced a recurring class of false positive alerts within 60 days of joining.
- Contributed to Angular dashboard consuming GraphQL APIs to surface inline transaction metadata for fraud analysts, eliminating dependency on a secondary internal system per case review and cutting average investigation time from 8 min to 5 min.
- Optimized PostgreSQL queries on one of the highest traffic transaction lookup path via composite indexing, query plan analysis, and Redis caching; p95 latency dropped from 480ms to 325ms (32% improvement).
- Contributed to AWS EKS deployment configuration using Terraform-provisioned infrastructure; corrected liveness probe thresholds that were misconfigured for the service startup time, reducing pod restart incidents by 67% during peak market hours.

Quest Diagnostics | Secaucus, NJ | Healthcare | Jan 2025 - Nov 2025

Full Stack Engineer

- Diagnosed a performance failure on the lab results portal that only manifested for patients with 3+ years of test history; the results viewer was loading the complete historical dataset on every page open with no pagination or lazy loading at any layer, pushing payloads to 2-3MB on hospital networks. Rebuilt the data fetching strategy to load only the current visit summary on open with on-demand fetch for history; page load dropped from 4s to 1.5s (63% improvement).
- Designed Java 17/Spring Boot REST API microservices with a GraphQL query layer; eliminated 5 over-fetching endpoints and reduced average API payload size by 41%.
- Reduced Oracle DB patient metadata lookup from 160ms to 90ms (44%) via JPA query optimization, composite indexing, and Redis caching on high-frequency paths.
- Built automated test coverage across 6 critical lab ordering workflows that had none before, using JUnit 5/Mockito and Cypress via GitHub Actions and Azure DevOps; reduced regression escape rate to staging by 80%.
- Wrote Python utilities for internal lab data validation and HL7 message parsing, automating pre-processing of 10K+ daily lab result records before ingestion into the reporting pipeline.

Informatica | Hyderabad, India | Enterprise SaaS | Sep 2022 - Dec 2023

Java Backend Engineer | *Client: Citco*

- Investigated a lineage graph defect reported by an enterprise customer after a connector clone feature shipped; traced root cause to the clone operation copying an internal identifier used by the lineage deduplication service as a unique key, causing cloned connectors to render as duplicate nodes with conflicting edges. Fixed the clone operation to generate a new unique identifier and added a database constraint that should have caught the bad write at the source.

- Contributed to the IDMC connector metadata service storing configuration templates for 200+ data connectors consumed by multiple internal teams; reduced service response time from 120ms to 80ms by rewriting N+1 query patterns and adding composite indexes across 3 high frequency endpoints.
- Built the IDMC metadata configuration UI in React.js/TypeScript using Factory and Observer patterns, giving admins self-serve control over 8+ microservice configurations and eliminating a 2-3 hour manual change request process.
- Migrated synchronous metadata sync to a Kafka event-driven pipeline; decoupled 5 downstream consumers and resolved timeout failures blocking UI provisioning under peak load.
- Built a Python service for internal metadata validation and transformation, ingesting payloads from 5 upstream systems, applying schema validation, and publishing clean events to Kafka; replaced a manual QA step that took 4+ hours per release cycle.

Bharti Airtel | Hyderabad, India | Telecom | May 2021 - Aug 2022
Software Engineer

- Diagnosed a SIM provisioning failure that spiked during flash sales but was absent under normal load; traced it to a HTTP client library version bump during a Java 11 migration that introduced a 5-second default connection timeout the previous version did not have. Postpaid activations required one additional downstream call that pushed total call time over the new threshold under load. Fixed by setting the timeout explicitly while the team determined the correct long-term value.
- Built Spring Boot REST APIs for the subscriber provisioning service handling SIM activations, plan changes, and account updates; refactored monolithic billing module into 4 independent microservices, reducing deployment cycle from bi-weekly to per-service and cutting critical billing defect rate by 40%.
- Implemented Spring Security authentication and RBAC across customer and internal portals for three user tiers.
- Optimized 4 high frequency MySQL billing queries via N+1 ORM rewriting and targeted indexing; maintained 87% unit and integration test coverage on core billing modules using JUnit 5/Mockito.
- Automated subscriber data reconciliation between CRM and billing systems using Python, eliminating a weekly manual sync that required 5+ hours of engineering time and produced roughly 200 mismatched records per run.

TECHNICAL SKILLS

Languages: Java (17/21), Python, TypeScript, JavaScript, SQL, HTML5, CSS3, C++
Backend: Spring Boot 3.x, Spring MVC, Spring Security, Spring Data JPA, Hibernate, FastAPI, Flask, REST API Design, Microservices Architecture, Event-Driven Architecture
Frontend: React.js, Angular, TypeScript, Tailwind CSS, Bootstrap, Vite
Messaging and APIs: Apache Kafka, RabbitMQ, GraphQL, gRPC, Protobuf, REST, OpenAPI/Swagger, Resilience4j
Databases: PostgreSQL, MySQL, Oracle DB, Redis, MongoDB
Cloud and DevOps: AWS (EKS, EC2, S3, RDS, Lambda), Azure (AKS, App Service), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, Azure DevOps, Maven, Gradle
Monitoring and Testing: Prometheus, Grafana, ELK Stack, JUnit 5, Mockito, Pytest, Cypress, Selenium, RestAssured, Postman, Distributed Tracing
Security: OAuth 2.0, JWT, OpenID Connect, LDAP, RBAC, Spring Security
Python Ecosystem: FastAPI, Flask, Pandas, NumPy, Pydantic, SQLAlchemy, Pytest, data pipeline scripting, automation scripting
Practices: Agile/Scrum, System Design, OOP, SOLID Principles, Design Patterns (Factory, Builder, Singleton, Observer, Strategy), Code Review, CI/CD, TDD

EDUCATION

Rowan University | M.S. Computer Science | Jan 2024 - Dec 2025 | GPA 3.9 | New Jersey
Relevant coursework: Distributed Systems, Advanced Algorithms, Cloud Computing, Database Systems, Software Engineering, Scalability and System Design

CERTIFICATIONS

- Oracle Certified Professional, Java SE 11 Developer | Oracle | 2023
- Oracle Certified Associate, Java SE 8 Programmer | Oracle | 2022